RESPONSE TO REFEREE #1

Dear Referee,

Thank you for your comments and suggestions. We have carefully considered your feedback and have revised the paper accordingly.

MAJOR COMMENTS I

Most important, I think the authors could try and illustrate a little bit more the novel empirical implications of their model, ideally by comparing the model and the data. In particular, what empirical implications does your model have that others don't (e.g. using survey data)? I find the quantitative section a little disappointing in that regard, since you match mostly standard moments. The facts presented in section 2, while useful, are fairly well known and general.

- *One simple implication that comes to mind is that the model predicts that earnings (or dividends, in the model) become more volatile when the share of wealth held by optimistic is larger, e.g. when the expansion is longer. Do we have evidence for this? Even something highly stylized would help.*
- *The model is about TFP growth, but some of the empirics use earnings growth, Some of the empirics use dividend growth. Do you think it would make sense to look at TFP forecasts (or GDP forecasts perhaps)? These changed a lot over time, for instance in the 1990s. There is a literature on how long it takes to learn about TFP shocks; for instance, Edge Laubach Williams JME 2007.*
- *Could we measure the dispersion in beliefs in IBES (or the SPF, or...) and see if the dispersion has implications? I guess in the model it has to be wealth-weighted.*
- *Some models that might be natural comparisons are the ones where agents are heterogeneous in risk aversion; when beliefs are different but fixed; and when there is not full risk sharing between the worker and the entrepreneur (which I think can be done relatively easily assuming for instance hand to mouth workers).*
- *I apologize that this comment is a little vague, as I don't have a very clear suggestion, but hopefully you understand my desire for more empirical implications.*

We thank you for this thoughtful comment. In the revised version of the paper, we substantially expanded the discussion of the model's novel empirical implications and added new theoretical and empirical material to better illustrate what distinguishes our framework from related models.

1. State-dependent link between belief dispersion and trading volume. We focus in particular on your suggestion regarding belief dispersion. The revised paper develops a new empirical implication of the model: under rank-alternating beliefs, the relationship between belief dispersion and trading volume is state-dependent. Specifically, turnover responds more strongly to disagreement in recessions than in expansions.

This asymmetry arises from the interaction of two forces in the model: (i) a rebalancing effect driven by wealth dynamics, and (ii) a change-in-beliefs effect that operates when agents switch from optimism to pessimism (or vice versa) across states. Under rank-alternating beliefs, these forces reinforce each other in downturns but partially offset each other in expansions, generating a sharp and testable prediction.

We added a new subsection that formally characterizes this mechanism and derives closed-form expressions linking turnover to belief dispersion. The section then takes this prediction to the data. Using IBES analyst expectations, we construct a measure of aggregate disagreement that mirrors the model's structure and estimate time-series regressions of stock market turnover on this measure.

Consistent with the theory, we find that turnover is significantly more sensitive to disagreement during recessions, with the strongest effects occurring at the onset of downturns. The results are robust to controlling for aggregate volatility (VIX) and market returns.

To the best of our knowledge, this state-dependent link between belief dispersion and trading volume has not been documented in the literature and constitutes a novel empirical implication of our framework.

Section 5.3 of the revised paper provides a detailed discussion. The first part derives the analytical relationship between turnover and belief dispersion. The second part implements the empirical test using IBES and CRSP data.

We reproduce below the theoretical discussion from Section 5.3 of the revised paper.

> Our belief taxonomy also has sharp implications for trading dynamics. As is standard in models with belief heterogeneity, trading volume is closely related to the degree of disagreement among investors. We show that not only the level but also the *type* of disagreement—as classified in Definition 3—plays a central role in shaping the dynamics of trading volume over the business cycle.

0.0.1 Analytical characterization of turnover

We adopt *share turnover* as our measure of trading volume, following Lo and Wang (2000), defined as

$$\tau_t \equiv \frac{1}{2}\sum_{i=1}^{I} \mu_i \left|S_{i,t} - S_{i,t-1}\right| = \frac{1}{2}\sum_{i=1}^{I} \left|\omega_{i,t}\eta_{i,t} - \omega_{i,t-1}\eta_{i,t-1}\right|,$$

where we divide by two to avoid double counting trades, and the second equality uses $\mu_i S_{i,t} = \omega_{i,t}\eta_{i,t}$.

In the absence of belief heterogeneity, $\omega_{i,t} = 1$ and $\eta_{i,t} = \eta_{i,t-1}$, so $\tau_t = 0$. With heterogeneous beliefs, turnover depends on the extent of belief dispersion. To characterize this dependence analytically, we consider a small deviation from a homogeneous-beliefs benchmark. Specifically, assume

$$p^i_{ss'} = p^*_{ss'} + \delta^i_{ss'}\epsilon, \qquad \delta^i_{sH} + \delta^i_{sL} = 0,$$

where $p^*_{ss'}$ denotes the common belief in the benchmark and ϵ controls the degree of dispersion. For simplicity, we take the benchmark belief to be symmetric and i.i.d., $p^*_{ss'} = \frac{1}{2}$. Appendix A.10 discusses the general case.

In this neighborhood, the portfolio share and next-period wealth share of investor i admit the expansions

$$\omega_i(X,s) = 1 + \kappa_\omega\left[p^i_{sH} - \overline{p}^m_{sH}(X,s)\right] + \mathcal{O}(\epsilon^2), \qquad \eta'_i(X,s,s') = \eta_i + \eta_i\frac{p^i_{ss'} - \overline{p}^m_{ss'}(X,s)}{p^*_{ss'}} + \mathcal{O}(\epsilon^2),$$

where $\kappa_\omega > 0$ and $\overline{p}^m_{ss'}(X,s) \equiv \sum_{j=1}^{I}\eta_j p^j_{ss'}$ denotes market beliefs. These expressions highlight how differences in beliefs drive portfolios and wealth dynamics.

To understand the determinants of trading volume, it is useful to decompose individual trading into two distinct components. The following lemma expresses an investor's net purchases as the sum of a rebalancing motive and a relative belief-updating motive.

Lemma 1. *Investor i's net purchases of shares are given by*

$$\mu_i \Delta S_i(X,s,s';\epsilon) = \underbrace{\Delta\eta_i(X,s,s')}_{\textit{rebalancing effect}} + \underbrace{\Delta\omega_i(X,s,s')\,\eta_i}_{\textit{change-in-beliefs effect}} + \mathcal{O}(\epsilon^2), \tag{1}$$

where

$$\Delta\eta_i(X,s,s') \equiv \eta_i\frac{p^i_{ss'} - \overline{p}^m_{ss'}(X,s)}{p^*_{ss'}}, \qquad \Delta\omega_i(X,s,s') \equiv \kappa_\omega\left[\left(p^i_{s'H} - \overline{p}^m_{s'H}(X,s)\right) - \left(p^i_{sH} - \overline{p}^m_{sH}(X,s)\right)\right].$$

Lemma 1 shows that individual trading behavior is governed by two distinct forces. The *rebalancing effect* captures trades required to maintain a constant portfolio share following changes in relative wealth. For example, when an optimistic investor gains wealth after a favorable shock, she must purchase additional shares to keep her portfolio composition unchanged. The *change-in-beliefs effect* reflects active portfolio reallocation driven by revisions in relative beliefs across states: if an investor becomes more optimistic in state s' than in state s, she increases her exposure to the risky asset upon the transition.

Whether these two effects reinforce or offset each other depends on how beliefs evolve across states. This interaction is the key mechanism behind the state-dependent relationship between belief dispersion and trading volume.

Turnover and belief taxonomy. As discussed in Section 3, we can express beliefs in terms of a persistence parameter, ρ_i, and a bias term, Δ_i:

$$p^i_{HH} = \frac{1+\rho_i}{2} + \Delta_i, \qquad p^i_{LH} = \frac{1-\rho_i}{2} + \Delta_i,$$

which implies that the conditional expectation of productivity growth is given by:

$$\mathbb{E}_{i,t}[x_{t+1}] = \overline{x}_i + \rho_i(x_t - \overline{x}_i), \tag{2}$$

where $\overline{x}_i$ is the subjective unconditional mean of x_t.

Consider first the case of rank-preserving beliefs, $\rho_i = \rho$, where investors differ on the level of optimism, Δ_i. In this case, the change-in-beliefs effect vanishes and turnover is driven solely by the rebalancing effect:

$$\tau(X, s, s'; \epsilon) = \sum_{i=1}^{I} \eta_i \left| p^i_{ss'} - \overline{p}^m_{ss'}(X, s) \right| + \mathcal{O}(\epsilon^2).$$

In this case, turnover is proportional to the average absolute deviation of beliefs, a natural measure of belief dispersion, so trading volume increases with disagreement.

Consider next the case of rank-alternating beliefs, $\Delta_i = \Delta$, where investors differ on the persistence parameter, ρ_i. In this case, the change-in-beliefs effect emerges when the economy switches states, $s \neq s'$, and it may either amplify or dampen the rebalancing effect. For instance, suppose the economy switches from H to L. Investors who were relatively optimistic in the boom lose wealth on impact and, at the same time, become relatively pessimistic in the downturn. Both forces push toward selling, so turnover rises sharply. By contrast, when the economy switches from L to H, the same investors become relatively optimistic and wish to raise their risky share, while

rebalancing incentives work in the opposite direction because their wealth share decreases. Turnover is therefore more sensitive to disagreement in recessions than in expansions.

The next proposition formalizes this intuition and delivers the central result of this section: trading volume is increasing in belief dispersion, and this relationship is asymmetric over the business cycle under rank-alternating beliefs.

Proposition 1. *Suppose investors' beliefs are rank-alternating, that is, they are given by* (2) *with* $\Delta_i = \Delta$. *Turnover as the economy switches from* s *to* s' *is given by*

$$\tau(X, s, s'; \epsilon) = \frac{\zeta(s, s')}{2} \sum_{i=1}^{I} \eta_i \left|\rho_i - \rho(X)\right| + \mathcal{O}(\epsilon^2),$$

where $\rho(X) \equiv \sum_{i=1}^{I} \eta_i \rho_i$, *and*

$$\zeta(s, s') \equiv \begin{cases} \kappa_\omega + 1, & \text{if } s = H \text{ and } s' = L, \\ \left|\kappa_\omega - 1\right|, & \text{if } s = L \text{ and } s' = H, \\ 1, & \text{if } s = s'. \end{cases}$$

The key message from Proposition 1 is that turnover increases with belief dispersion and, furthermore, that this sensitivity is stronger in downturns. The assumption of rank-alternating beliefs is essential for generating this asymmetry. With rank-preserving beliefs—investors are relatively optimistic or pessimistic by the same amount in both states—the change-in-beliefs effect is absent and turnover is driven primarily by rebalancing, so disagreement does not generate a disproportionately strong response of trading volume in bad times. Therefore, rank-alternating beliefs are key to capturing the state-dependent dynamics of stock market turnover.

Next, we reproduce the empirical discussion from Section 5.3 of the revised paper.

We now test the model's prediction that trading volume increases with belief dispersion and that this sensitivity is stronger during downturns.

Measuring disagreement and turnover. The model links trading dynamics to disagreement about aggregate shocks. To construct an empirical analogue, we combine analyst forecasts from I/B/E/S with stock market turnover from CRSP.

Let $\mathbb{E}_{j,t}[\Delta e_{i,t+1}]$ denote analyst j's forecast of firm i's earnings growth at time t, and let Δe_t denote realized aggregate earnings growth. To isolate the analyst-specific component reflecting beliefs about aggregate dynamics, we estimate

$$\mathbb{E}_{j,t}[\Delta e_{i,t+1}] = \alpha_{i,t} + \beta_{0,j} + \beta_{1,j} \Delta e_t + u_{j,i,t}, \tag{3}$$

where $\alpha_{i,t}$ is a firm–time fixed effect, $\beta_{0,j}$ is an analyst-specific intercept, $\beta_{1,j}$ captures the analyst's sensitivity to aggregate earnings growth, and $u_{j,i,t}$ is an error term.

The firm–time fixed effect absorbs firm-level information common to all analysts. The remaining analyst-specific terms therefore capture systematic differences in beliefs about aggregate earnings dynamics. We interpret the implied aggregate expectation of analyst j as

$$\mathbb{E}_{j,t}[\Delta e_{t+1}] = \beta_{0,j} + \beta_{1,j}\Delta e_t,$$

which mirrors the model's specification of expected productivity growth in Equation (2).

We then construct a time-varying *disagreement index* as the cross-sectional dispersion of these implied aggregate expectations:

$$\mathrm{DI}_t = \sigma_t\left[\beta_{0,j} + \beta_{1,j}\Delta e_t\right], \tag{4}$$

where $\sigma_t[\cdot]$ denotes the cross-sectional standard deviation at time t.

Even if analyst-specific coefficients are constant over time, disagreement varies with Δe_t. This reflects the model's mechanism that dispersion in persistence beliefs interacts with the current state of the economy.

To measure trading activity, we follow Lo and Wang (2000) and compute value-weighted stock market turnover from CRSP. Details on construction are provided in Appendix O1.1.

Figure 6 plots the time series of the disagreement index and turnover. Shaded areas correspond to NBER recessions. Disagreement rises sharply in downturns, periods in which turnover is also elevated.

Regression evidence. We formally assess the relationship using time-series regressions of turnover on the standardized disagreement index DI_t. Table 3 reports Newey–West standard errors with four lags.

Column (1) shows a positive but statistically insignificant association between turnover and disagreement. Column (2) adds an NBER recession indicator and its interaction with disagreement. The interaction term is positive and highly significant, indicating that turnover becomes substantially more sensitive to disagreement in recessions. The implied marginal effect of disagreement during recessions is economically meaningful: a one-standard-deviation increase in DI_t is associated with an increase in turnover of approximately 0.23 (the sum of the main and interaction coefficients), a sizeable fraction of its sample mean.

Column (3) shows that the interaction effect remains substantial when the recession dummy is excluded, indicating that the result is driven by differences in slopes rather than by shifts in the intercept.

Table 1: Turnover Regressions with Standardized Disagreement

	Baseline	Rec × DI	Rec × DI	Early Rec.	Early Rec.
	(1)	(2)	(3)	(4)	(5)
Disagreement (z-score)	0.075	0.054	0.054	0.069	0.069
	(0.047)	(0.039)	(0.039)	(0.047)	(0.047)
Recession		0.051			
		(0.048)			
Disagreement × Recession		0.176***	0.188***		
		(0.058)	(0.061)		
Early Rec. (first 2 qtrs)				0.014	
				(0.035)	
Disagreement × Early Rec.				0.095*	0.098*
				(0.051)	(0.051)
Marginal effect of DI in recession		0.230	0.242	0.164	0.167
Constant	0.311***	0.301***	0.305***	0.309***	0.309***
	(0.023)	(0.022)	(0.020)	(0.023)	(0.022)
Observations	158	158	158	158	158
R^2	0.212	0.352	0.344	0.235	0.235

Sample period: 1982Q2–2021Q3. Disagreement is standardized within sample. Newey–West (lag 4) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The model emphasizes the role of state transitions in generating this asymmetry. To proxy for transition periods, Columns (4) and (5) focus on the first two quarters of NBER recessions. Although the number of such observations is smaller, the interaction between disagreement and the transition indicator remains positive and statistically significant. This evidence is consistent with the model's prediction that turnover responds most strongly to disagreement at the onset of downturns.

The results are robust to controlling for aggregate market volatility (VIX) and market returns; these specifications are reported in Appendix O1.1.

Overall, the evidence supports the model's central implication: while trading volume increases with belief dispersion, this sensitivity is substantially stronger in recessions, particularly around state transitions.

2. TFP growth survey expectations. You raise the important question of whether the empirical analysis should focus more directly on expectations of TFP growth, given that the model is formulated in terms of productivity dynamics.

Direct survey data on TFP growth expectations are unfortunately limited.[1] How-

[1]For instance, Edge et al. (2007) use expectations about output per worker rather than TFP itself

ever, it is possible to construct survey-based proxies for productivity growth using expectations of GDP and employment growth from the Survey of Professional Forecasters (SPF).

In the revised paper, we therefore added a new subsection (Section 6.1) that evaluates whether the model can reproduce the empirical properties of subjective productivity-growth expectations.

Specifically, we estimate Coibion–Gorodnichenko (CG) regressions for two survey-based productivity measures—labor productivity growth and a TFP proxy—and compare the forecast-revision coefficient in the data to the corresponding coefficient obtained from simulated data in the calibrated model.

In the data, the CG coefficient is -0.63 for labor productivity and -0.50 for the TFP proxy, indicating substantial over-extrapolation in productivity expectations. The calibrated model implies a coefficient of -0.695. This value is quantitatively close to the empirical estimate for labor productivity and lies within one standard error of the estimate for the TFP proxy.

Importantly, the model was not calibrated to match productivity-expectation moments. Nevertheless, it reproduces the magnitude of the forecast-bias pattern observed in survey data. This shows that the belief distortions that generate endogenous movements in labor demand and risk premia also imply empirically realistic biases in productivity-growth expectations.

We believe this addition strengthens the paper by linking the model more directly to survey-based evidence on expectation formation. We thank you for this suggestion.

We reproduce below the discussion from Section 6.1 of the revised paper.

In the model, investors hold distorted beliefs about productivity growth. These distortions propagate to other variables — such as dividends and returns — through the likelihood-ratio representation:

$$\mathbb{E}_{i,t}[Z_{t+1}] = \mathbb{E}_t[\ell_{i,t+1} Z_{t+1}], \tag{5}$$

where $\ell_{i,t+1}$ is the likelihood-ratio between subjective and objective beliefs. Hence, a single distortion in productivity growth beliefs induces distortions in expectations of all equilibrium objects.[a]

It is therefore important to verify that the model is consistent with the empirical properties of subjective productivity growth expectations.

due to data limitations.

Coibion–Gorodnichenko regressions. We assess this using Coibion–Gorodnichenko (CG) regressions (Coibion and Gorodnichenko, 2015), which test for systematic forecast biases by regressing forecast errors on forecast revisions:

$$\Delta \mathrm{prod}_{t+h} - \mathbb{E}_{i,t}[\Delta \mathrm{prod}_{t+h}] = c + \beta_{CG}\left(\mathbb{E}_{i,t}[\Delta \mathrm{prod}_{t+h}] - \mathbb{E}_{i,t-1}[\Delta \mathrm{prod}_{t+h}]\right) + u_{t+h}. \quad (6)$$

Under full-information rational expectations, $\beta_{CG} = 0$. A negative estimate indicates over-extrapolation: agents revise forecasts too strongly in the direction of recent news.

Empirical implementation. As direct survey data on TFP growth expectations are limited, we consider two alternative measures of productivity growth using quarterly data from the Survey of Professional Forecasters (SPF):

$$\Delta \mathrm{prod}^{LP}_{t+1} = \Delta \hat{y}_{t+1} - \Delta \hat{n}_{t+1}, \quad (7)$$
$$\Delta \mathrm{prod}^{TFP}_{t+1} = \Delta \hat{y}_{t+1} - \alpha\, \Delta \hat{n}_{t+1}, \quad (8)$$

where $\Delta \hat{y}$ is GDP growth, $\Delta \hat{n}$ is employment growth, and we set $\alpha = 2/3$.

The first measure corresponds to labor productivity growth, and the second is a TFP proxy consistent with how TFP growth is defined in the model. Table 4 reports the results. For labor productivity, we estimate $\hat{\beta}_{CG} = -0.63$, and for the TFP proxy, $\hat{\beta}_{CG} = -0.50$, both statistically significant.

Model comparison. We perform the same CG regression using simulated data from the calibrated model. The model implies

$$\hat{\beta}^{\mathrm{model}}_{CG} = -0.695.$$

This value is very close to the empirical estimate for labor productivity and lies within one standard error of the estimate for the TFP proxy. These results support the model's mechanism. The same distortions that drive endogenous movements in labor demand and risk premia also generate empirically realistic forecast biases in productivity growth.

Overall, even though the model is not directly calibrated to productivity-expectations moments, it reproduces the key bias pattern observed in the data.

[a] Cui et al. (2024) estimate an empirical counterpart to the likelihood ratio and show that it explains a large fraction of cross-sectional variation in survey expectations.

3. Volatility amplification and the share of wealth held by optimists. The model also implies that dividend volatility increases when the share of wealth held by op-

Table 2: Coibion–Gorodnichenko Regressions for Productivity Measures

	Labor Productivity	TFP Proxy
Forecast revision coefficient β_{CG}	-0.63***	-0.50**
	(0.030)	(0.198)
Constant	-0.15	-0.25
	(0.289)	(0.259)
Observations	86	86
R^2	0.3697	0.0579
Adj. R^2	0.3622	0.0467
Sample	2004Q1–2025Q2	2004Q1–2025Q2

Notes: Each column reports OLS estimates of $FE_t = c + \beta\, Revision_t + u_t$, where $h = 1$ quarter. Reported standard errors (in parentheses) are Newey–West HAC with lag 4. Labor productivity uses $\Delta y - \Delta n$. TFP proxy uses $\Delta y - \alpha \Delta n$ with $\alpha = 2/3$. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (based on HAC p-values).

timistic investors is larger—for instance, following a prolonged expansion. In the revised paper, we clarify this mechanism and make explicit that the amplification operates through endogenous labor demand and operating leverage.

The mechanism unfolds in three steps. First, when wealth becomes more concentrated in the hands of optimistic investors, the overall level of market optimism rises and objective risk premia decline. Optimistic investors are more willing to bear risk, which compresses the risk premium, as shown in Section 5.1.

Second, lower risk premia increase the present value of future profits and induce firms to expand labor demand. This raises the labor share of output.

Third, a higher labor share increases operating leverage: because labor is chosen in advance and represents a quasi-fixed cost, profits (and hence dividends) become more sensitive to productivity shocks. As a result, dividend volatility increases when optimism—and therefore labor demand—is high.

Directly measuring wealth-weighted optimism in the data is challenging. However, each component of this mechanism is supported by existing empirical evidence. First, the link between optimism and lower expected returns is well documented (e.g., De La O and Myers 2021). Second, higher labor share predicts lower expected returns, consistent with compressed risk premia (Santos and Veronesi 2006), and aggregate hiring rates negatively predict stock returns (Belo et al. 2023), consistent with the labor-demand channel. Third, Donangelo et al. (2019) show that the labor share is a proxy for operating leverage and that higher labor share is associated with greater sensitivity of profits to shocks.

Taken together, this body of evidence is consistent with the model's prediction

that periods of elevated optimism—when wealth is more concentrated among optimists— are associated with greater operating leverage and higher dividend volatility.

We make the connection with the empirical evidence in the discussion of the model's predictions in Section 4 of the revised paper. The discussion of this mechanism is reproduced below.

Frothy markets and risk build-ups. An implication of Corollary 2 is that the risk premium declines with the length of economic expansions. To the extent that risk premia are reflected in credit spreads, the model is consistent with the discussion in López-Salido et al. (2017) and Krishnamurthy and Muir (2017) that economic booms are characterized by "frothy" market conditions driven by credit-market sentiment.

Importantly, the model delivers a joint prediction: as optimism builds during prolonged expansions, objective risk premia decline while dividend volatility rises. The compression in spreads reflects the greater risk-bearing capacity of optimistic investors. At the same time, lower risk premia raise labor demand, increase the labor share, and amplify operating leverage. Because labor is chosen in advance, profits, and therefore dividends, become more sensitive to productivity shocks. This combination implies an endogenous *risk build-up* during booms: fragility increases precisely when spreads are low.

As emphasized by Krishnamurthy and Li (2020), generating simultaneously low spreads and rising risk exposure is challenging in standard macro-finance models, such as He and Krishnamurthy (2013) and Brunnermeier and Sannikov (2014). In contrast, heterogeneous extrapolative beliefs naturally generate this pattern in our framework.

Each link in this mechanism is supported by existing evidence. First, survey-based measures of optimism are associated with lower expected returns, consistent with compressed risk premia (e.g., De La O and Myers 2021). Second, measures of labor demand and labor share predict lower future stock returns (Santos and Veronesi 2006; Belo et al. 2023), consistent with the link between labor demand and risk premia. Third, higher labor share is associated with stronger operating leverage and greater sensitivity of profits to shocks (Donangelo et al. 2019). Taken together, these findings support the interpretation that periods of elevated optimism are characterized by compressed spreads and increased fragility.

4. Distinction from alternative mechanisms. Finally, we clarify in the revised draft how our predictions differ from alternative heterogeneous-agent frameworks.

You mention two natural benchmarks: models with heterogeneous risk aversion, and models where there is not full risk sharing between workers and investors.

Models with heterogeneous risk aversion implies the ranking of investors in terms of risk-taking is fixed: more risk-tolerant agents tend to take more risk. This property is analogous to the case of rank-preserving beliefs shown in Figure 4. Hence, the implications of models with heterogeneous risk aversion should be similar to models with rank-preserving beliefs, and it will be different from models with rank-alternating beliefs.

We discuss this point in more detail in Section 5.2 of the revised paper.

Discussion: belief heterogeneity vs. risk aversion. As discussed above, rank-alternating beliefs amplify inherent economic fluctuations. Therefore, not only do differences in investors' risk appetite matter for business cycles, but also differences in their propensity to take risks. For instance, models of heterogeneous risk aversion can generate portfolio dispersion and time variation in expected returns. However, the ranking of agents in terms of risk-taking is always the same. This property is analogous to the case of rank-preserving beliefs shown in Figure 4. In contrast, there is no counterpart to the rank-alternating beliefs with heterogeneous risk aversion, as the agent taking less risk in booms is also taking more risk in the bust.

You also mention models where there is not full risk sharing between workers and investors. Appendix B describes a version of the model where labor is supplied by hand-to-mouth workers rather than by investors. We show that this modification leaves our main results essentially unchanged. In particular, in the special case of linear labor disutility ($\nu = 0$) considered in Section 4, the equilibrium allocation and asset prices are identical to those in the baseline model.

We reproduce below the characterization of the equilibrium in the hand-to-mouth version of the model from Appendix B.

[Missing quote `HandToMouthWorkers`]

We refer to this analysis in Section 4 of the revised paper.

Households choose both portfolios and labor supply. In general, this joint choice makes closed-form characterizations difficult. However, under complete markets and GHH preferences, an *as-if* result applies: the household problem can be recast as a pure portfolio problem in an endowment economy, abstracting from labor decisions.[a]

[a] In Appendix B, we show our results also hold in an environment where labor is supplied by hand-to-mouth workers rather than by investors.

Taking stock. In response to your comment, the revised paper now makes explicit several empirical implications of the model that go beyond matching standard asset-pricing moments. In particular, we show that the framework can account for:

1. A state-dependent relationship between belief dispersion and trading volume.
2. The magnitude of forecast biases in productivity-growth expectations, as measured by Coibion–Gorodnichenko regressions.
3. The joint occurrence of low risk premia and elevated risk exposure during booms.
4. The link between survey expectations and objective risk premia.
5. Standard asset-pricing moments, including the level of the equity premium, return volatility, and return predictability.
6. The cyclicality of subjective risk premia and the relationship between labor income and expected returns.

While the model remains stylized, these results illustrate that heterogeneous extrapolative beliefs generate a set of testable implications across asset prices, survey expectations, trading activity, and labor market dynamics.

We are grateful for your comment, which prompted us to clarify and expand the empirical content of the paper. We believe the revised version provides a more complete and transparent assessment of the model's empirical relevance.

Major comments II

> *It wasn't fully clear to me how much of the dynamics in the model are due to changes in average beliefs, versus the risk channel emphasized by the author. That is, even without risk aversion, more optimistic beliefs would lead to more hiring (as figure 3 shows). Perhaps this point is worth showing analytically and/or quantitatively.*

Thank you for this insightful comment. In the revised paper, we clarify the respective roles of average beliefs, risk premia, and risk aversion, both analytically and quantitatively.

First, we explicitly distinguish between *subjective* and *objective* risk premia. While the subjective risk premium depends primarily on perceived risk (second moments), the objective risk premium also responds to distortions in first moments induced by belief-driven price movements. As a result, waves of optimism substantially compress the objective risk premium while having a much more muted effect on the subjective risk premium. We make this distinction explicit in Section 5 and illustrate it in Figure 2.

Second, because labor demand is tightly linked to the objective risk premium through the risk-neutral expectation of productivity growth, movements in average (market) beliefs are the primary driver of hiring dynamics in the model. This clarifies

why optimism leads to higher employment even when subjective risk perceptions change little.

Third, risk aversion plays a crucial supporting role. It ensures well-defined interior portfolio choices and guarantees that investors agree on state prices, allowing for a well-defined firm objective. In addition, risk aversion is essential for the wealth dynamics underlying internal propagation: without it, optimistic investors would concentrate wealth in good states and be wiped out following adverse realizations, and the amplification mechanism would break down.

We now make these points explicit in the analytical section and reinforce them in the quantitative analysis, where we show that the model generates large movements in objective risk premia and labor demand driven by shifts in market beliefs, alongside relatively muted movements in subjective risk premia, consistent with the empirical evidence.

As I noted above, the result 3 (joint determination of hiring and asset prices, i.e. feedback loop), which is quite interesting, is not clearly demonstrated or explained; in particular its quantitative significance or (again) empirical implication. My understanding is that this should work through the EZ sdf. When agents are in aggregate optimistic, there is more risk taking, and higher risk taking today leads to more volatility tomorrow in profits. How does that feed back into today's risk premia? Because of the complete markets, consumption tomorrow is not actually more volatile, even if there is more risk. (If it were, it would create more volatility in the SDF, which would dampen the effect, actually, rather than amplify it.) So I am a little bit at loss to understand that. It would be great to explain this better and illustrate it in the model (perhaps adding a column of table 3).

Thank you for this comment. In the revised paper, we substantially expand the discussion of why hiring and asset prices are jointly determined and clarify the precise mechanism through which beliefs affect risk premia and real outcomes.

First, we add a new segment that isolates the role of the hiring friction. We show that if firms are allowed to choose labor after productivity is realized, a form of *macro–finance separation* obtains: fluctuations in market beliefs affect asset prices but have no real effects on employment or output. This result holds regardless of preferences. We introduce this new result immediately after the characterization of the risk-neutral expectation of productivity growth:

The real effects of market beliefs. We have seen that, in our baseline economy, asset prices and labor demand are jointly determined. The timing of hiring decisions is crucial for this result. To make this point explicit, the next proposition shows that market beliefs have no real effects when firms are allowed to adjust labor after productivity

is realized.

Proposition 2 (Real effects of market beliefs)**.** *Suppose firms choose hours after observing current productivity. Then hours, detrended profits, and the return on the risky asset are independent of market beliefs:*

$$h(X,s) = \left(\frac{\alpha x_s}{\xi}\right)^{\frac{1}{1-\alpha}}, \qquad \pi(X,s) = (1-\alpha)\left(\frac{\alpha}{\xi}\right)^{\frac{\alpha}{1-\alpha}} x_s^{\frac{1}{1-\alpha}}, \qquad R_r(X,s,s') = \frac{x_s}{\beta}\left(\frac{x_{s'}}{x_s}\right)^{\frac{1}{1-\alpha}}. \quad (9)$$

Fluctuations in market beliefs affect the risk-free rate and the risk premium, but they do not affect real variables or the expected return and volatility of the risky asset under the objective measure.

Proposition 2 establishes a form of *macro–finance separation* (Tallarini, 2000): in the absence of a hiring friction, belief-driven fluctuations in risk premia have asset-pricing implications but no real effects on employment or output.

The timing assumption on hiring breaks this separation. When labor must be chosen in advance, hiring is risky from the firm's perspective. An increase in optimism compresses the risk premium, raises the value of risky assets, and—through the firm's optimality condition—leads to higher employment. In this case, asset prices and labor demand are simultaneously determined.

Absent fluctuations in market beliefs, employment would be constant in this economy, consistent with the labor volatility puzzle documented by Shimer (2010). Section 6 explores the quantitative implications of this mechanism by assessing how much labor volatility the model can generate under empirically realistic fluctuations in beliefs.

This comparison makes clear that the joint determination of asset prices and labor demand does not rely on Epstein–Zin preferences per se. Even with log utility, hiring and asset prices are jointly determined as long as labor must be chosen in advance. Conversely, if labor is flexible, separation holds even under Epstein–Zin preferences.

Second, regarding quantitative significance, Section 6 shows that in the absence of belief fluctuations the model generates no employment volatility, replicating the labor volatility puzzle of Shimer (2010). We then show that realistically sized belief fluctuations generate substantial and empirically plausible movements in both asset prices and employment, allowing the model to jointly account for the equity premium, equity volatility, and labor volatility puzzles.

Third, we clarify that consumption does respond to beliefs once the macro–finance separation is broken. In particular, shifts in beliefs changes the volatility of *net consumption* (consumption minus labor disutility) via the operating leverage channel. Given the non-separability of consumption and labor in preferences, it is net consumption —rather than consumption itself— that enters the SDF.

Finally, we explain how belief-driven optimism feeds back into current risk pre-

mia even under complete markets. We show that, while the subjective risk premium depends only on perceived risk (second moments), the objective risk premium also responds to distortions in first moments induced by belief-driven price movements. We make this mechanism explicit in the following segment:

Beliefs and risk premia. Fluctuations in market beliefs generate movements in both risk premia and labor demand. Why does optimism lead to changes in risk premia?

To answer this question, it is useful to distinguish between *subjective* and *objective* risk premia. With a representative investor, the subjective risk premium is simply given by:

$$\mathbb{E}_{i,s}\left[R_r^e(X,s,s')\right] \approx \mathrm{Var}_{i,s}\left[R_r^e(X,s,s')\right]. \tag{10}$$

Thus, absent changes in perceived volatility, shifts in beliefs about average productivity growth have little effect on the subjective risk premium in this case.

In contrast, the objective risk premium responds to differences in first moments. Using the equilibrium expressions derived above, it can be written as

$$\mathbb{E}_s\left[R_r^e(X,s,s')\right] = \mathbb{E}_{i,s}\left[R_r^e(X,s,s')\right] - \frac{\mathbb{E}_{i,s}[x_{s'}] - \mathbb{E}_s[x_{s'}]}{(1-\alpha)\mathcal{L}'(X,s)}. \tag{11}$$

When investors become optimistic, they expect higher future dividends and bid up asset prices. From the perspective of a rational investor, higher prices imply lower expected returns going forward, compressing the objective risk premium.

Figure 2 illustrates this mechanism. An increase in optimism, captured by a rise in market beliefs $\overline{p}_{sH}^{m}(X,s)$, substantially reduces the objective risk premium, while having a more muted effect on the subjective risk premium. This pattern is consistent with empirical evidence documenting large movements in expected returns with relatively stable subjective risk perceptions (see, e.g., De La O and Myers 2021 and Nagel and Xu 2023).

We hope these additions clarify both the logic and the quantitative relevance of the feedback loop between beliefs, asset prices, and labor demand.

Another theoretical point which confuses me has to do with the impact of the length of the expansion (or, alternatively, the share of wealth of the optimists) on the amplification conditional on a switch in the Markov state. I can see how if the expansion is longer, more wealth is in the hand of the optimists. However the change of their wealth if a switch happens depends also on what positions they take. If, as the expansion lengthens, they take more and more risky positions, they might become less wealthy following a Markov switch (than if the switch had happened early on). I am guessing the model assumption (preferences?) prevents this, but it

might be worth explaining. Up to you. (Perhaps explaining the portfolios more would help.)

Thank you for this question. This is a central mechanism in the paper, and we agree that its logic should be fully transparent. In the revised paper, we expand the discussion of the amplification mechanism and add a new paragraph explaining the role of risk aversion and portfolio choice.

The result that longer expansions lead to larger downturns under rank-alternating beliefs relies on two key properties of the wealth dynamics. First, the wealth share of optimistic investors increases during booms and decreases during recessions. Second, as implied by Proposition 5, an investor's future wealth share is increasing in her current wealth share. As you conjecture, preferences play a crucial role in ensuring these properties.

In the log-utility case, these results follow directly. Appendix A.8 shows that they extend to unit-EIS economies with risk-averse investors. By contrast, the mechanism breaks down with risk-neutral investors: in that case, optimists concentrate their portfolios in the good state and are effectively wiped out following adverse realizations (Geanakoplos, 2010), so the wealth accumulated during the expansion does not carry over to downturns.

More generally, as long as investors are risk-averse, they optimally distribute wealth across states, ensuring that optimistic investors retain wealth after a switch. The degree of risk aversion governs the strength of the mechanism: when investors are close to risk-neutral, amplification is attenuated, while higher risk aversion strengthens the effect. We make this role of risk aversion explicit in a new paragraph in Section 5:

The role of risk aversion. The results above rely on two key features of the wealth dynamics. First, the wealth share of optimistic investors increases during booms and decreases during recessions. Second, as implied by Proposition 5, the future wealth share of any investor is increasing in her current wealth share:

$$\frac{\partial \eta_i'(X, s, s')}{\partial \eta_i} = \frac{p^i_{ss'}}{\overline{p}^m_{ss'}(X, s)} > 0. \tag{12}$$

This property contrasts with models featuring risk-neutral investors, in which optimistic agents are driven to concentrate their wealth in the good state and are wiped out in the bad state (Geanakoplos, 2010). With risk-averse investors, by contrast, agents choose to allocate wealth across both states of the world. As a result, optimistic investors retain wealth in downturns, allowing the belief-driven amplification mechanism to operate.

The strength of this mechanism depends on the degree of risk aversion. When optimistic investors are close to risk-neutral, additional wealth accumulated during a boom is largely tilted toward the good state, attenuating the amplification of fluctuations in bad times. Higher risk aversion induces investors to distribute wealth more evenly across states, increasing the influence of optimistic investors during downturns and thereby strengthening the amplification of adverse shocks.

I find it really hard to go through the paper (despite its elegance) because the authors' notation and presentation is not very clear. This is a little frustrating because the authors seem to favor some notation beauty, which is not what the reader needs. I think the presentation could be much improved. There are two dimensions to improve:

1. *First, many results are presented with a lot of algebra but not a lot of intuition. I can see you like that you can solve everything in closed form, and this is great, but the value to the reader is to show what the formula says. Otherwise you should put it in appendix... For instance Prop 2 requires more comments; Corollary 2 as well; etc.*

Thank you for this comment. In the revised paper, we have made a concerted effort to improve the intuition and clarity surrounding the main analytical results, while keeping closed-form expressions only where they help highlight the underlying economic mechanisms.

Regarding Proposition 3, the previous version presented the closed-form expressions with limited guidance. In the revision, we reorganize the discussion to make the logic transparent from the outset. We first lay out three standard implications of log utility: (i) the consumption–wealth ratio and price–dividend ratio are constant, (ii) the risky return is proportional to dividend growth, and (iii) the risk-free rate equals the risk-neutral expectation of the risky return:

Asset prices. Under log utility, both the consumption–wealth ratio and the price–dividend ratio are constant:

$$c_i(X, s) = 1 - \beta, \qquad q(X, s) = \frac{\beta}{1 - \beta}, \tag{13}$$

which follows directly from Equations (13) and (26). As a result, the return on the risky asset is proportional to dividend growth:

$$R_{r,t+1} = \frac{1 + q_{t+1}}{q_t} \frac{\pi_{t+1}}{\pi_t} = \beta^{-1} \frac{\pi_{t+1}}{\pi_t}.$$

Using the expression for dividend growth derived in Equation (22), the gross re-

turn on the risky asset can be written as

$$R_r(X, s, s') = \beta^{-1} x_s \frac{x_{s'} - \alpha \mathcal{L}'(X, s)}{x_s - \alpha \mathcal{L}} \left(\frac{\mathcal{L}'(X, s)}{\mathcal{L}} \right)^{\frac{\alpha}{1-\alpha}}.$$

The risk-free rate equals the risk-neutral expectation of the return on the risky asset. Using the fact that $\mathcal{L}'(X, s)$ is the risk-neutral expectation of $x_{s'}$, we obtain

$$R_f(X, s) = \beta^{-1} x_s \frac{(1-\alpha)\mathcal{L}'(X, s)}{x_s - \alpha \mathcal{L}} \left(\frac{\mathcal{L}'(X, s)}{\mathcal{L}} \right)^{\frac{\alpha}{1-\alpha}}.$$

Proposition 3 then follows immediately from these building blocks:

Proposition 3 (Risk-free rate and risk premium)**.** *Given $\mathcal{L}'(X, s)$:*

(i) The risk-free rate satisfies

$$R_f(X, s) = (1-\alpha) \frac{x_s}{\beta} \frac{\mathcal{L}'(X, s)^{\frac{1}{1-\alpha}}}{x_s \mathcal{L}^{\frac{\alpha}{1-\alpha}} - \alpha \mathcal{L}^{\frac{1}{1-\alpha}}}. \tag{14}$$

(ii) The conditional risk premium is

$$\mathbb{E}_s\left[R_r^e(X, s, s')\right] = \frac{1}{1-\alpha} \frac{\mathbb{E}_s[x_{s'}] - \mathcal{L}'(X, s)}{\mathcal{L}'(X, s)}, \tag{15}$$

which is decreasing in $\mathcal{L}'(X, s)$.

We also add a dedicated discussion explaining the economic implications of the result, emphasizing how the risk-free rate and the risk premium depend on the risk-neutral expectation of productivity growth:

Proposition 3 shows that asset prices are fully pinned down by the current state (X, s) and the risk-neutral expectation of productivity growth, $\mathcal{L}'(X, s)$. Intuitively, a higher $\mathcal{L}'(X, s)$ reflects either higher expected productivity growth or lower aggregate risk, both of which raise the risk-free rate.

At the same time, interest rates are decreasing in current productivity growth x_s. As discussed in Section 4, the timing of hiring induces mean reversion in dividends: following a negative productivity shock, expected dividend growth is high, raising equilibrium interest rates.

Finally, the risk premium is increasing in the wedge between physical and risk-neutral expectations of productivity growth. Conditional on $\mathbb{E}_s[x_{s'}]$, a lower $\mathcal{L}'(X, s)$ corresponds to a larger risk adjustment and therefore a higher expected excess return.

A similar intuition-first approach is now used throughout Section 5, including for Corollary 1.

Regarding Corollary 2, we now provide a more detailed explanation of the economic intuition behind the result, including a discussion of the role of risk aversion, as mentioned in the answer to your previous question.

We hope these changes make the analytical results easier to follow and clarify what the closed-form expressions imply economically.

2. *Second, the notation and presentation could be a lot clearer. In particular, the state space should be made explicit from the beginning and used consistently, rather than introducing an abstract* X *and later substituting it with other objects (e.g.,* L *or LDF). I also found the terminology "LDF" confusing, and the large number of return concepts difficult to track.*

Thank you for this suggestion. In the revised paper, we have substantially simplified both notation and exposition, with the goal of making the state space and economic mechanisms transparent from the outset.

First, following your recommendation, we now make the state space explicit at the beginning of Section 4. The first paragraph of that section now reads:

We now present a recursive characterization of the equilibrium. The economy evolves with an exogenous state s and an aggregate endogenous state X. In our setting, X consists of the wealth distribution (i.e., the wealth shares of investors) and, given the timing of labor decisions, an aggregate variable that pins down firms' hiring choices. As we show below, this variable corresponds to the risk-neutral expectation of productivity growth.

In particular, we clearly state that the aggregate state consists of the wealth distribution and an aggregate variable that pins down firms' hiring decisions, which we subsequently show corresponds to the risk-neutral expectation of productivity growth.

Second, we have removed the term "LDF" entirely and instead use "risk-neutral expectation of productivity growth" throughout the paper. We agree that "LDF" is potentially confusing, while the new terminology is both standard and economically transparent. This change is reflected consistently in the revised paper. For example, in the firm's problem we now define $\mathcal{L}_t$ as:

The term $\mathcal{L}_t$ corresponds to the *risk-neutral expectation of productivity growth*:

$$\mathcal{L}_t = \mathbb{E}_t\left[\frac{\Lambda_{t,t+1}}{\mathbb{E}_t[\Lambda_{t,t+1}]}\, x_{t+1}\right].$$

Productivity growth is evaluated under probabilities proportional to $p_{ss'}\Lambda(X, s, s')$, that is, under risk-neutral probabilities. Hence, the firm's optimality condition implies

a zero expected labor wedge under these risk-adjusted expectations.

Third, to address the proliferation of return concepts, we now focus in Sections 4 and 5 on a special case of the model with linear labor disutility. This case preserves the role of labor and the main economic mechanisms, but eliminates human wealth and substantially reduces the number of distinct return objects that need to be tracked.[2] In this special case, stocks are the only risky asset, and we use standard portfolio-choice notation, with $R_{r,t}$ denoting the return on the risky asset and $R_{f,t}$ the risk-free rate. We return to the general case in Section 6.

We hope these changes make the notation more uniform and the exposition easier to follow.

MINOR COMMENTS I

1. *The eqn at the bottom of p14 surprised me – isn't it possible that the denominator be negative? (that should be impossible, o/w the SDF would be <0). But I don't see at this point why a negative growth couldn't generate a positive return*

Thank you for raising this point. You are correct that the expression used in the paper implicitly assumed that the denominator is positive. While this assumption holds in some special cases (e.g., under log utility), it does not hold in general. We have therefore revised the paper to remove this implicit restriction.

General argument. The equation in question follows directly from inverting the no-arbitrage conditions for stocks and bonds. Solving the no-arbitrage conditions for the SDF yields

$$\Lambda(X, s, L) = \frac{R_r^e(X, s, H)}{p_{sL}\, \Delta R_r(X, s)}, \qquad \Lambda(X, s, H) = -\frac{R_r^e(X, s, L)}{p_{sH}\, \Delta R_r(X, s)},$$

where R_r^e denotes excess returns and $\Delta R_r(X, s) \equiv R_r(X, s, H) - R_r(X, s, L)$. These expressions are exact implications of no-arbitrage and hold for all parameter values.

Importantly, no-arbitrage requires that excess returns cannot have the same sign in all states. Hence, either $R_r^e(X, s, H) > 0 > R_r^e(X, s, L)$ or the reverse, implying that the sign of $\Delta R_r(X, s)$ is not fixed a priori. When $\Delta R_r(X, s) < 0$, the SDF remains strictly positive, but the compact expression used in the paper must be adjusted accordingly.

[2] In the original setting, the fact that human wealth was non-zero led us to keep track of two different risky assets: stocks and the surplus claim. Under the special case of linear labor disutility, human wealth is zero, and the surplus claim is equal to the stock, substantially simplifying the exposition.

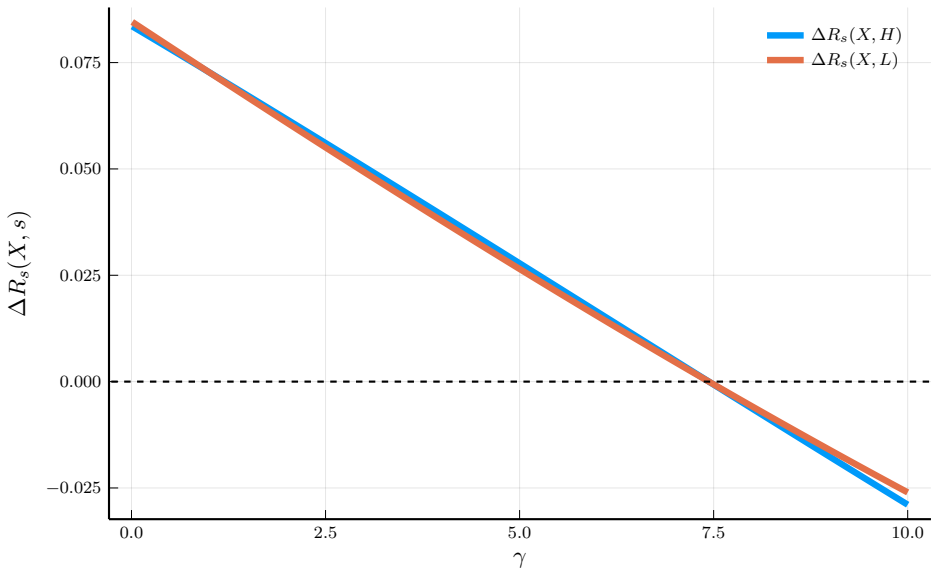

Figure 1: Mehra-Prescott limit

Note: The solution is computed with the parameters $\beta = 0.99$, $x_H = 1.018 + 0.036$, $x_L = 1.018 - 0.036$, $p_{HH} = p_{LL} = (1+\rho)/2$, $\rho = 0.15$.

Possibility of $\Delta R_r(X,s) < 0$. To show that $\Delta R_r(X,s) < 0$ can indeed arise in equilibrium, we consider the Mehra–Prescott limit (CRRA preferences, no belief heterogeneity, and vanishing labor income). In this case, for sufficiently high risk aversion (low EIS), the high-productivity state is associated with a high interest rate, which can reduce asset prices enough to generate lower realized returns despite higher cash flows. Figure 1 illustrates this possibility.

Revision to the paper. Motivated by this observation, we have revised the paper to avoid assuming that $\Delta R_r(X,s)$ is positive and to state the SDF expressions in a form that is valid for all parameter configurations, as shown below:

Given objective probabilities, the economy-wide stochastic discount factor can be recovered from asset prices by inverting the no-arbitrage conditions:

$$\Lambda(X,s,L) = \frac{1}{p_{sL}} \frac{R_r^e(X,s,H)}{\Delta R_r(X,s)}, \qquad \Lambda(X,s,H) = -\frac{1}{p_{sH}} \frac{R_r^e(X,s,L)}{\Delta R_r(X,s)}, \tag{16}$$

where $R_r^e(X,s,s') \equiv R_r(X,s,s')/R_f(X,s) - 1$ denotes the excess return on the risky asset and $\Delta R_r(X,s) \equiv R_r(X,s,H) - R_r(X,s,L)$ is the difference in realized returns across states.

2. *I like Figures 4 and 5 a lot, but the numbers are crazy. Perhaps show the figures for EZ utility instead of IES=RA=1 so as to get something more ballpark?*

Thanks for this suggestion. We agree that, in the previous version, the numerical values in Figures 4 and 5 were far from realistic, which could distract from the economic mechanism the figures are meant to illustrate.

Because the purpose of these figures is to provide intuition within the analytically transparent log-utility case, we retain the assumption of log utility. However, we have recalibrated the model so that it matches standard asset-pricing moments more closely. In particular, the revised calibration generates reasonable values for the Sharpe ratio, which was notably off in the earlier version.

This adjustment preserves the simplicity of the log-utility environment while ensuring that the numerical magnitudes shown in the figures are more in line with empirical benchmarks.

3. *The data in table 3 is taken from three different papers – with in some cases mutually exclusive samples! This is not serious. It won't change anything, but you should do a better job of choosing a sample and measuring yourself these easily available variables.*

We agree. In the previous submission, the data column in the (then) Table 3 relied on moments reported in multiple papers with different (and in some cases mutually exclusive) samples. In the revised paper, we now compute *all* data moments from scratch using a single, transparent set of sources and clearly stated sample windows. In particular, we construct interest rates, excess returns, and dividend growth from the Shiller historical market dataset; consumption growth from BEA/FRED nondurables-and-services consumption per capita; and hours volatility from CPS series accessed via the BLS API (see the Data Appendix, Section O1.3).

In addition, beyond point estimates, we now quantify sampling uncertainty by computing moving-block bootstrap standard errors (and corresponding ± 2 s.e. intervals) for each moment, and we report these intervals directly in the updated Table 5 (formerly Table 3). Details of the bootstrap procedure are described in the Supplementary Appendix (Data Appendix, Section O1.3).

4. *Table 3 would benefit from clear column names.*

We thank you for this suggestion. In the revised paper, we have added clear column labels: "Endowment" (column 1, the endowment limit), "Production" (column 2, the production economy with labor friction), "Homogeneous" and "Heterogeneous" (columns 3–4, subjective beliefs without and with heterogeneity), "High

elast." (column 5, higher labor supply elasticity), and "Data" (column 6). We also expand the table notes to describe the exact specification considered in each column.

> 5. *I am not sure if the label of "diagnostic expectations" (DE) is correct for your Markov chain assumption. That's because DE, as you describe them, rely on an overreaction of beliefs to a shock. But suppose that in your setup, the Markov state does not change. That's a shock — the realization of TFP growth differs from its conditional expectation — but it does not affect beliefs. For this reason, I am not sure that I would call this DE or even extrapolative. It seems to me more precise to describe the setup as capturing different biases in beliefs across consumers, both their average belief bias and how the bias depends on the aggregate state.*

Thank you for this thoughtful comment. We agree that the interpretation of *diagnostic expectations* depends critically on what is taken to be the relevant *reference group*, and that this distinction is especially salient when comparing discrete-state and continuous-state environments. Our view is that the label *diagnostic expectations* is appropriate for our Markov-chain setup under the natural choice of reference group in that context. Below, we clarify this interpretation and explain why it differs from the continuous-state case that motivates your intuition.

Diagnostic expectations. Following Gennaioli and Shleifer (2010), the *representativeness* of a trait $T = t$ for group G relative to a reference group $-G$ is defined as

$$R(t, G, -G) \equiv \frac{\Pr(T = t \mid G)}{\Pr(T = t \mid -G)}. \tag{17}$$

A trait is said to be *diagnostic* of a group if it is more representative of that group than of the reference group.

As in Bordalo et al. (2016), agents distort objective probabilities according to

$$\Pr^i(T = t \mid G) = \frac{\Pr(T = t \mid G)\, h^i\big(R(t, G, -G)\big)}{\sum_{t' \in \mathcal{T}} \Pr(T = t' \mid G)\, h^i\big(R(t', G, -G)\big)}, \tag{18}$$

where $h^i(\cdot)$ is a weighting function. We refer to expectations formed under this distorted probability as *diagnostic expectations*.

A commonly used specification is $h^i(R) = R^{\theta_i}$, yielding

$$\Pr^i(T = t \mid G) = \underbrace{\Pr(T = t \mid G)}_{\text{rational beliefs}} \times \underbrace{\left(\frac{\Pr(T = t \mid G)}{\Pr(T = t \mid -G)}\right)^{\theta_i}}_{\text{belief distortion}} \times C, \tag{19}$$

where C is a normalizing constant.

Diagnostic expectations in a Markov-chain environment. In our model, this structure maps naturally to a discrete-state Markov process. In particular, the trait T corresponds to next period's state s_{t+1}, the group G corresponds to the current state $s_t = s$, and the reference group $-G$ corresponds to the alternative state $s_t = -s$. Under this mapping,

$$\overset{i}{\Pr}(s_{t+1} = s' \mid s_t = s) = p_{ss'} \left(\frac{p_{ss'}}{p_{-ss'}} \right)^{\theta_i} \times C, \tag{20}$$

which coincides with the belief distortion used in the paper.

If productivity growth is persistent under rational beliefs (i.e., $p_{HH} > p_{LH}$ and $p_{LL} > p_{HL}$), diagnostic expectations imply that agents overweight state persistence: conditional on being in a high (low) state, remaining in that state is perceived as especially representative. In this sense, agents overreact to the current state, which is the discrete-state analogue of extrapolation.

Why this differs from the continuous-state intuition. Your interpretation applies more directly to continuous-state processes, where a shock can be defined as a realization that differs from the conditional mean. In that setting, diagnostic expectations typically require specifying a reference group such as the expected value based on prior information.

For example, Bordalo et al. (2020) consider an AR(1) process,

$$x_{t+1} = \rho x_t + \sigma \varepsilon_{t+1}, \tag{21}$$

and define the reference group using the conditional expectation ρx_{t-1} before x_t is realized. The distorted density then compares likelihoods relative to this benchmark, generating overreaction to unexpected realizations.

In contrast, with a binary Markov chain, there is a natural reference group—the alternative state. As a result, belief distortions operate through state-dependent representativeness rather than through deviations from a conditional mean. This explains why beliefs do not change following a realization that does not trigger a state transition: representativeness is defined at the level of states, not innovations.

Conclusion. In sum, our use of the term *diagnostic expectations* is consistent with the formal definition in the literature once the appropriate reference group is specified for a discrete-state environment. Intuitively, when productivity is persistent, the current state becomes especially representative under diagnostic beliefs, leading agents to overestimate persistence. This mechanism is closely related to extrapolation, though implemented differently than in continuous-state models.

To avoid confusion, we have added a clarifying footnote in the paper explicitly discussing how our definition maps into standard formulations of diagnostic expectations, as shown below:

Diagnostic expectations correspond to the following belief structure:[a]

$$p^i_{sH} = \underbrace{p_{sH}}_{\text{rational beliefs}} \times \underbrace{\left(\frac{p_{sH}}{p_{-sH}}\right)^{\theta_i}}_{\text{belief distortion}} \times C,$$

[a]Diagnostic expectations are defined using distorted probabilities of the form $\Pr^i(T = t \mid G) \propto \Pr(T = t \mid G)\left(\frac{\Pr(T=t|G)}{\Pr(T=t|-G)}\right)^{\theta_i}$ (see, e.g., Bordalo et al. 2018). In our setting, the trait T is next period's state s', the group G is the current state s, and the reference group $-G$ is the alternative state $-s$.

Of course, if you or the editor still feel that the terminology risks confusion, we would be happy to adopt alternative wording.

MINOR COMMENTS II

1. *As explained above, I think the use of LDF is confusing rather than its more logical name, expected productivity growth under* Q.

Thank you for this suggestion. In the revised paper, we have removed the term "LDF" entirely and consistently use *risk-neutral productivity growth* throughout. We agree that this terminology is more natural and transparent, and it clarifies the economic interpretation of the object as an expectation under the risk-neutral measure.

2. *Footnote 19 is quite obscure.*

To avoid any confusion, we decided to remove the footnote.

3. *Some of the authors' language is a little peculiar. I am pretty sure that "intrapolation" is not widely used. (Googling it brings you directly to their paper...) What does it mean? "Interpolation"? Even that is not clear. It is worth defining what you mean by that (or even perhaps finding a more common term that captures the behavior you model).*

We thank you for this comment and agree. The term "intrapolative" is not standard and may cause confusion. In the revision, we replace it with *contrarian beliefs*, which more transparently captures the behavior we model —agents who expect reversals rather than persistence and are pessimistic in booms but optimistic in downturns. We believe this terminology is more consistent with the asset-pricing and expectations literature.

4. *Similarly, the paper oscillates between "operational leverage" and "operating leverage"; I think the latter is more widely used.*

Thank you for pointing out this typo. We have replaced "operational leverage" with "operating leverage" throughout the paper.

5. *I find the citation of Sargent at the end a little cringe – what value does the cite add. But up to you of course.*

Thank you for this comment. If you are comfortable with it, we would like to keep the citation. We view the Sargent citation as adding useful context: it highlights the motivation for departing from full-information rational expectations, from a leading scholar of the rational expectations revolution. Of course, if you prefer, we are happy to remove the citation.

6. *Succesfully typo in conclusion*

Fixed.

7. *We "define" no p12 – it's already defined.*

In the revised version of the paper, we do not discuss the human wealth at this point anymore, so this phrase has been removed.

8. *Ci0t tilde in eqn 9 p13? Why not just Ci tilde?*

Fixed.

9. *I kept getting confused with the subscripts* r, e *in returns (and I think the authors themselves get confused — see* $j \in \{r, b\}$ *after equation (11)). Why not use simpler notation, such as* R_s *for stock returns and* R_x *for excess returns?*

As discussed in our response to your second major comment, we have substantially simplified the notation in the revised paper by focusing on the case of linear labor disutility ($\nu = 0$). This special case preserves the main economic mechanisms while allowing for a much more transparent exposition.

In the revised version, there is a single risky asset (stocks). We denote its gross return by $R_{r,t}$ and its excess return by $R^e_{r,t} \equiv R_{r,t}/R_{f,t} - 1$, which follows standard asset-pricing notation (see, e.g., **?**). The additional return objects present in the previous version have been eliminated, and we have checked that the notation is now used consistently throughout the paper.

We hope this change resolves the confusion regarding return notation.

10. *"attenuate the subsequent decline in TFP growth" - I think you mean, attenuate the effects of the subsequent decline in TFP growth*

Fixed.

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